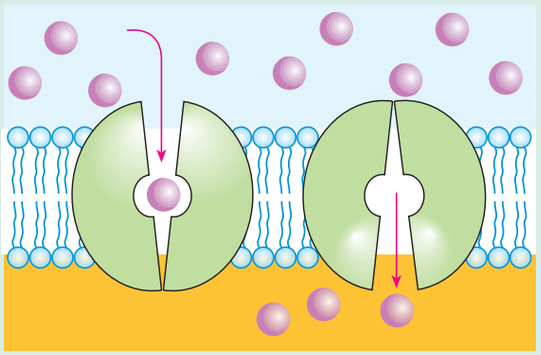
ii. Active transport - the movement of substances across the membrane using energy from adenosine triphosphate (ATP).

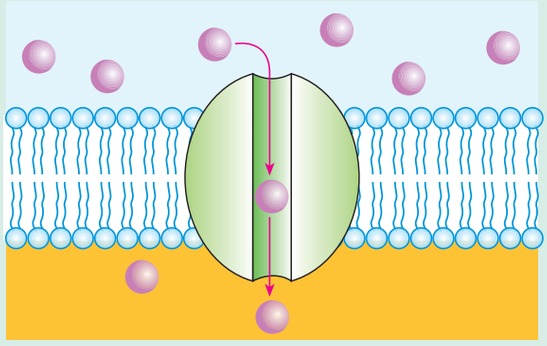
1.3.1.Passive transport

To appreciate the movements in the above mentioned passive mode, let us consider the terms viz., gradients and diffusion

A concentration gradient is the difference in concentration of a substance across a space. Molecules (or ions) will spread/diffuse from where they are more concentrated to where they are less concentrated until they are equally distributed in that space. (When molecules move in this way, they are said to move down their concentration gradient.) Three common types of passive transport include simple diffusion, osmosis, and facilitated diffusion.



Transport of matter with facilitated diffusion



Substance passage through carrier protein channel

Fig 5:. Facilitated Diffusion

1.3.2. Facilitated diffusion

Facilitated diffusion of substances crossing the cell (plasma) membrane takes place with the help of proteins such as channel proteins and carrier proteins. Channel proteins are less selective than carrier proteins, and usually mildly discriminate between their cargo based on size and charge. Carrier proteins are more selective, often only allowing one particular type of molecule only to cross.

1.3.3. Active Transport

For the transport methods described above, the cell expends no energy. Membrane proteins that aid in the passive transport of substances do so without the use of ATP. Active transport differs from diffusion in that molecules are transported away from thermodynamic equilibrium (hence, energy is required). The required energy can come from the hydrolysis of ATP, from electron movement, or from light. The energy moves a substance across a membrane, often with the help of protein carriers, and usually against its concentration gradient. One of the most common types of active transport involves proteins that serve as pumps. The word “pump” probably conjures up thoughts of using energy to pump up the tire of a bicycle or a basketball. Similarly, energy from ATP is required for these membrane proteins to transport substances—molecules or ions—across the membrane, usually against their concentration gradients (from an area of low concentration to an area of high concentration).

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